UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

PHYSICS – IMPORTANT FORMULAE & CONSTANTS

GRADE 11 ADVANCED | TERM 1 | 2026–2027

MODULE 8 | DESCRIBING ROTATIONAL MOTION

ANGULAR DISPLACEMENT

$$\Delta\theta = \theta_f - \theta_i$$

$\Delta\theta$ = angular displacement (rad); θ_i, θ_f = initial, final angular position (rad).

ARC LENGTH

$$s = r\Delta\theta$$

s = arc length (m); r = distance from the rotation axis (m). Use θ in radians.

AVERAGE ANGULAR VELOCITY

$$\bar{\omega} = \frac{\Delta\theta}{\Delta t}$$

$\bar{\omega}$ = average angular velocity (rad/s); Δt = elapsed time (s).

AVERAGE ANGULAR ACCELERATION

$$\bar{\alpha} = \frac{\omega_f - \omega_i}{\Delta t}$$

$\bar{\alpha}$ = angular acceleration (rad/s²); ω_i, ω_f = angular velocities (rad/s).

ANGULAR SPEED & FREQUENCY

$$\omega = 2\pi f = \frac{2\pi}{T}$$

f = frequency (Hz = s^{-1}); T = period (s); ω = angular speed (rad/s).

PERIOD & NUMBER OF TURNS

$$T = \frac{1}{f} \qquad |\Delta\theta| = 2\pi N$$

N = number of turns in one direction (unitless). One complete revolution is 2π rad.

TANGENTIAL SPEED

$$v = r\omega$$

v = tangential speed (m/s); r = radius (m); ω = angular speed (rad/s).

TANGENTIAL ACCELERATION

$$a_t = r\alpha$$

a_t = tangential acceleration (m/s²); α = angular acceleration (rad/s²).

MODULE 8 | ANGLE & SPEED CONVERSIONS

FROM	TO
Degrees → radians	θ(rad) = θ(°) × π/180; radians → degrees: multiply by 180/π.
Rotational speed	rev/s → rad/s: multiply by 2π; rev/min → rad/s: multiply by 2π/60.

MODULE 8 | TORQUE & STATIC EQUILIBRIUM

TORQUE

$$\tau = rF\sin\theta$$

τ = torque (N·m); r = pivot-to-force distance (m); F = force (N); θ = angle between r and F.

PERPENDICULAR LEVER ARM

$$r_\perp = r\sin\theta \qquad \tau = Fr_\perp$$

$r_\perp$ = shortest perpendicular distance from the pivot to the force's line of action (m).

PERPENDICULAR FORCE

$$\tau = rF \quad (\theta = 90^\circ)$$

Torque is greatest for a perpendicular force. A force through the pivot produces zero torque.

NET TORQUE

$$\tau_{\text{net}} = \sum\tau$$

Add torques with signs. Usually counterclockwise is positive and clockwise is negative.

TRANSLATIONAL EQUILIBRIUM

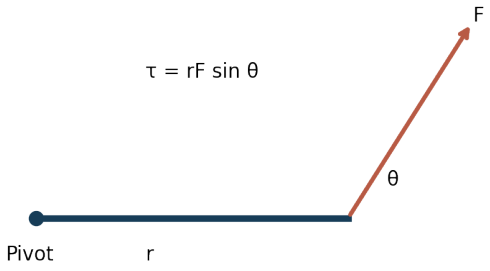
$$\sum F_x = 0 \quad \sum F_y = 0$$

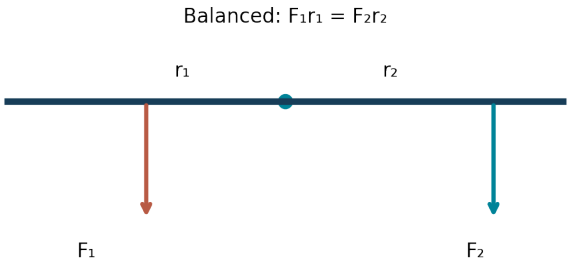
Net force is zero (N). There is no translational acceleration.

ROTATIONAL EQUILIBRIUM

$$\sum\tau = 0$$

Net torque is zero (N·m). For static equilibrium, both net force and net torque must vanish.





MODULE 8 | BALANCE & STABILITY

BALANCED LEVER / SEESAW

$$F_1 r_1 = F_2 r_2$$

For two perpendicular opposing torques: F = force (N); r = lever arm (m).

TWO MASSES ON A LIGHT SEESAW

$$m_1 r_1 = m_2 r_2$$

Use when weights are mg and g cancels. m = mass (kg); r = perpendicular distance (m).

Stable support: the vertical line through the centre of mass must fall within the base. A lower centre of mass and wider base increase stability.

MODULE 8 | MOMENT OF INERTIA — STATE THE AXIS

ONE POINT MASS

$$I = mr^2$$

I = moment of inertia (kg·m²); m = point mass (kg); r = distance from axis (m).

SEVERAL POINT MASSES

$$I_{\text{total}} = \sum m_j r_j^2$$

Add each mass's contribution about the same axis. Do not use distance between masses in place of r.

THIN HOOP

$$I = MR^2$$

M = total mass (kg); R = radius (m). Axis through the centre, perpendicular to the hoop.

SOLID UNIFORM CYLINDER / DISK

$$I = \frac{1}{2}MR^2$$

M = total mass (kg); R = radius (m). Axis along the cylinder's symmetry axis.

SOLID UNIFORM SPHERE

$$I = \frac{2}{5}MR^2$$

M = total mass (kg); R = radius (m). Axis through the centre.

UNIFORM THIN ROD — CENTRE

$$I = \frac{1}{12}M\ell^2$$

M = rod mass (kg); l = rod length (m). Axis through the midpoint, perpendicular to the rod.

MODULE 8 | ROTATIONAL DYNAMICS

UNIFORM THIN ROD — END

$$I = \frac{1}{3}M\ell^2$$

Axis through one end, perpendicular to the rod. The moment of inertia depends on the axis.

NEWTON'S SECOND LAW — ROTATION

$$\tau_{\text{net}} = I\alpha$$

τ_{net} = net torque (N·m); I = moment of inertia (kg·m²); α = angular acceleration (rad/s²).

REARRANGEMENT / CHECK	USE
Thin rectangular plate	I = ¹⁄₁₂M(l² + w²); axis through the centre, perpendicular to the plate.
Angular acceleration	$\alpha = \tau_{net}/I$; for constant α, $\omega_f = \omega_i + \alpha\Delta t$.
Find force at a rim	F = τ/(r sin θ); for a tangential force, F = τ/r.
Mass distribution	For a point mass, doubling its distance from the axis multiplies its I by 4.

MODULE 9 | LINEAR MOMENTUM, IMPULSE & COLLISIONS

LINEAR MOMENTUM

$$p = mv$$

p = momentum (kg·m/s); m = mass (kg); v = signed velocity (m/s).

CHANGE IN MOMENTUM

$$\Delta p = p_f - p_i = m(v_f - v_i)$$

For constant mass. Momentum is a vector: keep the direction signs.

IMPULSE–MOMENTUM THEOREM

$$J = F_{\text{avg}}\Delta t = \Delta p$$

J = impulse (N·s = kg·m/s); F_{avg} = average net force (N); Δt = contact time (s).

AVERAGE FORCE

$$F_{\text{avg}} = \frac{\Delta p}{\Delta t}$$

For the same Δp, a longer contact time gives a smaller average force.

SYSTEM MOMENTUM CONSERVED

$$\sum p_i = \sum p_f$$

Applies when the net external impulse on the system is zero or negligible.

TWO-BODY CONSERVATION

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

u = initial velocity; v = final velocity (m/s). Use the same positive direction for both objects.

OBJECTS STICK TOGETHER

$$v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

v = shared final velocity (m/s). This is a perfectly inelastic collision.

RECOIL FROM REST

$$m_1v_1 = -m_2v_2$$

Initial total momentum is zero. Final momenta are equal in magnitude and opposite in direction.

MODULE 9 | FORCE–TIME GRAPH

Impulse = signed area under the net force–time graph. Rectangle: FΔt; triangle: ½ × base × height. For a changing force, use the graph area or $F_{avg}\Delta t$.

RELATION	PHYSICAL MEANING
$J_1 = -J_2$	Two interacting objects exert equal and opposite forces over the same interval; their impulses are opposite.
$F_{ext,avg}\Delta t = \Delta p_{system}$	Internal forces cancel in the system sum; external impulse changes total momentum.

MODULE 9 | ANGULAR MOMENTUM & ANGULAR IMPULSE

ANGULAR MOMENTUM

$$L = I\omega$$

L = angular momentum (kg·m²/s); I = moment of inertia (kg·m²); ω = angular velocity (rad/s).

CHANGE IN ANGULAR MOMENTUM

$$\Delta L = L_f - L_i$$

Use signed angular momentum. If I changes, calculate Iω separately in each state.

ANGULAR IMPULSE

$$\tau_{\text{avg}}\Delta t = \Delta L$$

Angular impulse (N·m·s = kg·m²/s); τ_{avg} = average net external torque (N·m).

AVERAGE EXTERNAL TORQUE

$$\tau_{\text{avg}} = \frac{L_f - L_i}{\Delta t}$$

For constant I: $\tau_{avg} = I(\omega_f - \omega_i)/\Delta t$. Use Δt in seconds.

ANGULAR MOMENTUM CONSERVED

$$I_i\omega_i = I_f\omega_f$$

Use when net external torque about the chosen axis is zero or negligible.

CHANGING SHAPE / SPIN RATE

$$\omega_f = \frac{I_i\omega_i}{I_f}$$

Smaller I → larger |ω| when angular momentum is conserved, as in a skater pulling in the arms.

QUICK REFERENCE | SYMBOLS & SI UNITS

QUANTITY	SYMBOL	SI UNIT
Angular displacement	θ, Δθ	radian (rad)
Angular velocity / acceleration	ω / α	rad/s / rad/s²
Frequency / period	f / T	hertz (Hz) / second (s)
Force / torque	F / τ	newton (N) / newton metre (N·m)
Moment of inertia	I	kg·m²
Linear momentum / impulse	p / J	kg·m/s / N·s
Angular momentum / impulse	L / τΔt	kg·m²/s / N·m·s

Do not confuse: I is moment of inertia, l is rod length, and L is angular momentum. Choose a rotation axis and sign convention before substituting. Torque has unit N·m, although it is not energy.